# Sigma Transfer and Fiber Overlap: Pareto Transitions of the March-Graph Spectrum and the Sidon Fiber-Overlap Bound

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## 1 Abstract

We study the $\sigma$-spectrum $S(\sigma)$ of the march-graph program: the set of attainable $(|E|, |S|)$ classes at span $\sigma$ over the 5,872-shape universe of Sidon 9-shapes, and its Pareto frontier $\Pi(\sigma)$. The formalized content, all [PROVED] in Lean, is the *$\sigma$-transfer* suite (T5, T6, T7′, T8′) and the *fiber-overlap* suite. The transfer theorems: Pareto membership is down-set membership (Theorem 1, `mem_paretoMin_ iff`); every class's transition status across an adjacent span pair is a function of per-class down-set queries on the two spectra (Theorem 2, `status_local`: the T5 locality bar, old spectra only); the axioms A1–A3 impose no relation between consecutive spectra, witnessed by three explicit abstract models (Theorem 3, `no_forcing_equality`/`_disjoint`/`_sharing`: the T6 abstract-only bar); reflection duals of Pareto classes are never distinct Pareto classes (Theorem 4, `dual_emin`, `no_cotransition`, `A3_holds`). The fiber theorems: distinct pair-sum fibers of a Sidon set overlap in at most one point (Theorem 5, `fiber_overlap`, `fiber_overlap_identity`, `autocorr_le_one`), and top-interval sparsity holds conditionally on the [PROVED*] T1/T2 floors (Theorem 6, `top_interval_sparsity`). We also record, honestly, that Q-20's claimed sharp Pareto floor $|E| = 0 \implies |S| \geq 7$ is refuted by an exact six-hole witness (Proposition 1, [COMPUTED]).

## 2 Introduction

A Sidon set $A$ (a $B_2$ set) has all pair sums $a+b$ $(a, b \in A)$ essentially distinct. The march-graph program studies Sidon 9-shapes $A = \{0 = a_0 < \cdots < a_8 = \sigma\}$ of span $\sigma \leq 183$ through their *pair-sum shadow* $(D, H, W)$ and the $(E, S)$ hole counts (R264/R270/R273, cited as given). For each span $\sigma$ the *$\sigma$-spectrum* $S(\sigma)$ is the set of attainable $(|E|, |S|)$ classes over the 5,872-shape universe, and its $\leq$-minimal elements form the *Pareto frontier* $\Pi(\sigma)$ (195 distinct classes over 294 instances in the census, supplied datum). The program's transition analysis (R275) classifies, for each adjacent pair $\sigma \to \sigma+1$, how Pareto classes persist, demote, vanish, appear, or promote — the *transition DAG* further studied in the companion paper on transition DAGs.

This paper's formalized contribution is the abstract, machine-checked core of that analysis, kept strictly within what the Lean modules prove:

1. The down-set characterization of Pareto membership and the T5 transition-locality law (Theorems 1–2).
2. The T6 no-forcing theorem: three explicit abstract spectra show the floor axioms impose no relation between consecutive spectra (Theorem 3).
3. The T7′/T8′ dual statements for full-normalized-universe reflection (Theorem 4).

4. The fiber-overlap identity and bound, and the conditional top-interval sparsity law (Theorems 5, 6).

Every census number in this paper is [COMPUTED] (×2, two independent paths, exact agreement) and is never re-claimed as a theorem; every theorem is an abstract Lean-checked statement. The six-hole witness of Proposition 1 is a [COMPUTED] refutation we keep on the record.

# 3 Methods

## 3.1 Definitions

We use the R264/R270/R273 notation (supplied datum). Fix a Sidon 9-shape $A = \{0 = a_0 < \cdots < a_8 = \sigma\}$ of span $\sigma \le 183$.

- $P$ is the set of 45 pair sums $a + b$.
- $D = \bigcup_{s \in P}(s - A)$ is the union of the 45 *D-fibers* $f_s = s - A$.
- $H = \{s/2 : s \in P \text{ even}\}$ and $W = D \cup H$ $(A \subseteq D)$.
- $u = 183 - \sigma$; $J = [\sigma - 183, -1]$.
- $E = J \setminus D$ and $S = [0, \sigma] \setminus W$; $|G_{t_{\max}}| = |E| + |S|$.

$S(\sigma)$ is the $\sigma$-spectrum: the set of attainable $(|E|, |S|)$ classes at span $\sigma$ over the 5,872-shape universe. $\Pi(\sigma)$ is the Pareto frontier: the $\le$-minimal elements of $S(\sigma)$. In Lean: `paretoMin`$(S) = S$.filter$(\lambda p.\, \forall q \in S,\, q \le p \to q = p)$ and `downSet`$(S, c) = S$.filter$(\cdot \le c)$.

The R275 transition taxonomy (supplied datum) over an adjacent pair $\sigma \to \sigma + 1$ with spectra $S, S'$ assigns each class one of five statuses (Lean inductive `Status`): `persist`, `demoted`, `extinct`, `fresh`, `promoted`. In down-set form: persist $\iff c \in S' \wedge (\downarrow(c) \setminus \{c\}) \cap S' = \varnothing$; demoted $\iff c \in S' \wedge (\downarrow(c) \setminus \{c\}) \cap S' \neq \varnothing$; extinct $\iff c \notin S'$; promoted $\iff a \in S(\sigma) \setminus \Pi(\sigma)$ with blockers $\Pi(\sigma) \cap \downarrow(a) \neq \varnothing$; fresh $\iff a \notin S(\sigma)$.

The three floor axioms (Lean `A1`, `A2`, `A3`):

- A1 (floor): every class of $S$ has $e \ge \text{floor}$, $\text{floor}(\sigma) = 183 - 2\sigma$.
- A2 (reflection-closure): the reflected $e$-value appears at the same $s$.
- A3 (Pareto): $\Pi(S)$ is the $\le$-minimal set (definitional).

The T6 forcing models are three explicit abstract spectra with $m = \max(f(\sigma), f(\sigma + 1))$ and identity reflection: the *equality* model $S = S' = \{(m, 0)\}$; the *disjointness* model $S = \{(m, 0)\}$, $S' = \{(m, 1)\}$; the *sharing* model $S = S' = \{(m, 1), (m + 1, 0)\}$ (an antichain).

For the fiber suite: the *fiber* of $A$ at shift $s$ is $f_s = s - A = \{s - a : a \in A\}$ (Lean `fiber`$(A, s) = A$.image$(\lambda a.\, s - a)$). The *autocorrelation* of $A$ at $t$ is $|A \cap (A + t)|$. *Top-interval sparsity* is the bound $|D \cap [\sigma + 1, 183]| \le \sigma$ for $\sigma \le 91$, derived from T2's equality $|E_{\bar{A}}| = u - |D_A \cap [\sigma + 1, 183]|$ and T1's $|E_{\bar{A}}| \ge 183 - 2\sigma$ (both [PROVED*], cited with labels intact). Conjecture F fragments: F1: $|E| = 0 \implies |S| \ge 2$; F2: $|S| = 0 \implies |E| \ge 2$; Conjecture F: $|E| + |S| \ge 2$ for all Sidon 9-shapes.

Table 1: Formalized theorems, all [PROVED]. Module paths are repo-relative.

| Lean theorem | content | axioms |
|---|---|---|
| `Mathproofs/SigmaTrans.lean` | | |
| `mem_paretoMin_iff` | Pareto $\iff$ down-set singleton | standard |
| `status_local` | T5 transition locality | standard |
| `dag_deterministic` | statuses exhaustive/total | standard |
| `demoter_exists` | finite descent to Pareto | standard |
| `dual_emin` | T7′ dual $e$-min | standard |
| `no_cotransition` | T8′ no distinct co-Pareto dual | standard |
| `A3_holds` | Pareto axiom holds | standard |
| `no_forcing_equality` | T6 equality model | standard |
| `no_forcing_disjoint` | T6 disjoint model | standard |
| `no_forcing_sharing` | T6 sharing model | standard |
| `Mathproofs/FiberOverlap.lean` | | |
| `autocorr_le_one` | autocorrelation $\leq 1$ | standard |
| `fiber_overlap_identity` | overlap = autocorrelation | standard |
| `fiber_overlap` | fiber-overlap bound | standard |
| `f_iff_f1_f2` | F $\iff$ F1$\wedge$F2 (pure logic) | `[propext, Quot.sound]` |
| `top_interval_sparsity` | sparsity (conditional) | `[propext, Quot.sound]` |

### 3.2 Formalization note

The machine-checked content lives in the repository `derekblair/math-proofs`, Lean 4.34.1 with Mathlib pinned at commit `d13f23b`. Verification config: full-library `lake build` on the bakkie fleet machine (2026-10-11, 11:12 PDT): "Build completed successfully (17936 jobs)" — exit 0, 0 errors, 0 `sorry`s, 0 `admit`s in the log. The R310 label move took both modules from [PROVED*] to [PROVED] (charter rule 4), with `SigmaTrans` (Q-22) and `FiberOverlap` (Q-23) among the 8 queue modules built. Modules are md5-identical local $\leftrightarrow$ bakkie (`SigmaTrans.lean` `ab2398b4...`; `FiberOverlap.lean` `67b2a9a5...`), each registered exactly once per side. Axiom audits: `SigmaTrans` standard-only `[propext, Classical.choice, Quot.sound]`; `autocorr_le_one`, `fiber_overlap_identity`, `fiber_overlap`: `[propext, Classical.choice, Quot.sound]`; `f_iff_f1_f2`, `top_interval_sparsity`: `[propext, Quot.sound]`. Each module's header points at this paper's slug `papers/sigma-transfer-fiber-overlap-October-11-2026/`; this paper points back at the modules and the exact theorem names of Table 1.

The scope bar (Q-22, quoted scoped SURVIVES, 2026-10-10 21:31 PDT; Q-23, quoted SURVIVES, 2026-10-10 21:25 PDT) is part of the claim: T5 is *old-spectrum queries* only — each class's status is a function of per-class down-set queries on the two spectra; the census data itself stays [COMPUTED]. T6 is *abstract A1–A3 only* — the three models are Lean-checked existence statements over abstract finsets. T2′/T7′/T8′ are *full-normalized-universe* results with witness-level reflection (abstract reflection endomap with $s$-preservation); the per-shape T2′ content (`reflection_per_shape`, needing the $(S, E)$ evaluator formalized) is DEFERRED, documented in the module header, and not claimed here. The 103-pair transition census, the 558-case T5 down-set audit, the T6 model audits, the T2′/T7′/T8′ scripted audits, and the 5,813,280-pair fiber audit stay [COMPUTED]$\times$2.

## 4 Results

**Theorem 1** (Pareto membership; PROVED)**.** *For $S$ a finset of $\mathbb{N} \times \mathbb{N}$ and $c$ a class,*

$$c \in \texttt{paretoMin}(S) \iff c \in S \ \wedge\ \texttt{downSet}(S, c) = \{c\}. \tag{1}$$

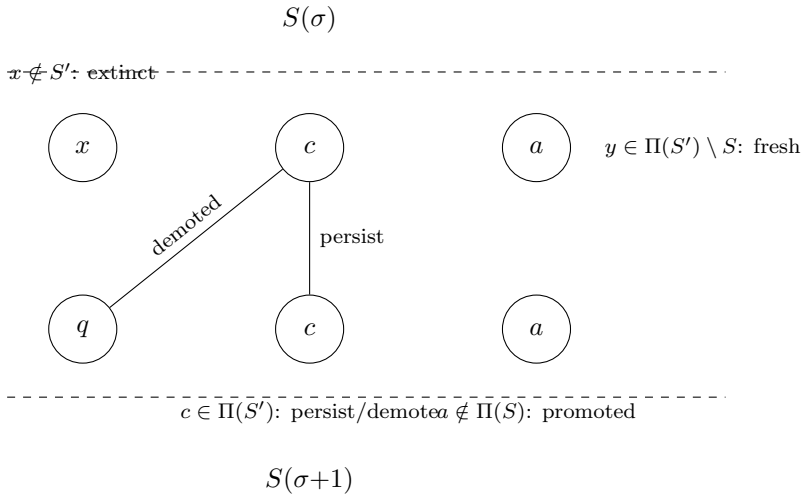



Figure 1: The T5 transition taxonomy (Theorem 2): a class's status across $S(\sigma) \to S(\sigma+1)$ is read off from per-class down-set queries on the two spectra. Arrows show persist ($c \in \Pi$ both sides) and demoted ($c$ lands on a dominating $q$); extinct, fresh, and promoted complete the five statuses.

**Theorem 2** (T5 transition locality; PROVED)**.** *The transition status of a class $c$ across adjacent spectra $S, S'$ is a function of per-class down-set queries on the two spectra:* `transStatus` *is defined by cases on $c \in S \wedge$* `downSet`$(S, c) = \{c\}$ *and the same predicate on $S'$, and* `dag_deterministic` *shows the five statuses are exhaustive and total over $\Pi(S) \cup \Pi(S')$. Moreover (*`demoter_exists`*), every non-Pareto class descends to a Pareto class: $c \in S$, $c \notin$* `paretoMin`$(S)$ *implies $\exists q \in$* `paretoMin`$(S)$ *with $q \leq c$, $q \neq c$.*

Theorem 2 is the entire content of T5 that we claim: the status function is *local* to the two spectra's down-set data. Nothing about the census's actual statuses is proved in Lean; the 558-record audit (below) is a supplied [COMPUTED] datum. The orientation of demote edges was additionally audited on the census (24 demote edges, 0 orientation errors, supplied datum).

**Theorem 3** (T6 no-forcing; PROVED)**.** *The axioms A1–A3 impose no relation between consecutive spectra. There exist abstract finsets satisfying A1–A3 at both spans with (i) identical Pareto minima (*`no_forcing_equality`*, $\{(m, 0)\}$ both sides), (ii) disjoint Pareto minima (*`no_forcing_disjoint`*, $\{(m, 0)\}$ vs $\{(m, 1)\}$), and (iii) an identical two-class antichain (*`no_forcing_sharing`*, $\{(m, 1), (m+1, 0)\}$ both sides, card 2), where $m = \max(f(\sigma), f(\sigma+1))$ and the reflection is the identity endomap.*

The claim is deliberately *existential and abstract*: the models are Lean-checked witnesses that the axioms do not force persist/demote/extinct/fresh/promote patterns. The census fact that 103/103 consecutive frontier sets are distinct is contingent and [COMPUTED], not a class-level law and not a theorem (supplied datum, R275 dossier). In particular, "nowhere $\sigma$-monotone is a frontier-set-level census fact, not a class-level law and not a theorem."

**Theorem 4** (T7′/T8′ dual statements; PROVED)**.** *Let $r$ be a reflection endomap on the class universe with s-preservation ($\forall p$, $(rp).2 = p.2$). Then (*`dual_emin`*): if $p \in$* `paretoMin`$(S)$ *and $rp \in S$, then $p.1 \leq (rp).1$; and (*`no_cotransition`*): if further $rp \in$* `paretoMin`$(S)$, *then $rp = p$ — the dual of a Pareto class is never a distinct Pareto class. Moreover (*`A3_holds`*) the Pareto axiom holds for all $S$.*

These are full-normalized-universe statements with witness-level reflection (Q-22 scoped bar): $\sigma$-internality is captured by $r$ being an endomap on the class universe. The census's T7′/T8′

Table 2: Transition census (supplied [COMPUTED]×2 datum, R275 dossier; two independent paths, exact agreement). None of these numbers is a theorem.

| quantity | value |
|---|---|
| shapes in universe | 5,872 |
| adjacent-pair transition records | 103 |
| distinct Pareto classes | 195 |
| Pareto-class instances | 294 |
| class-transition records | 558 |
| dominance-logic events (20 demoted + 13 promoted) | 33 (5.8%) |
| persist events (over 13 pairs; max share 2 at 86>87) | 14 |
| merges / splits | 3 / 4 |
| demote edges (orientation-audited) | 24 |
| extinct + fresh (sampling churn) | 259 + 266 |
| vanish total = appear total | 279 |

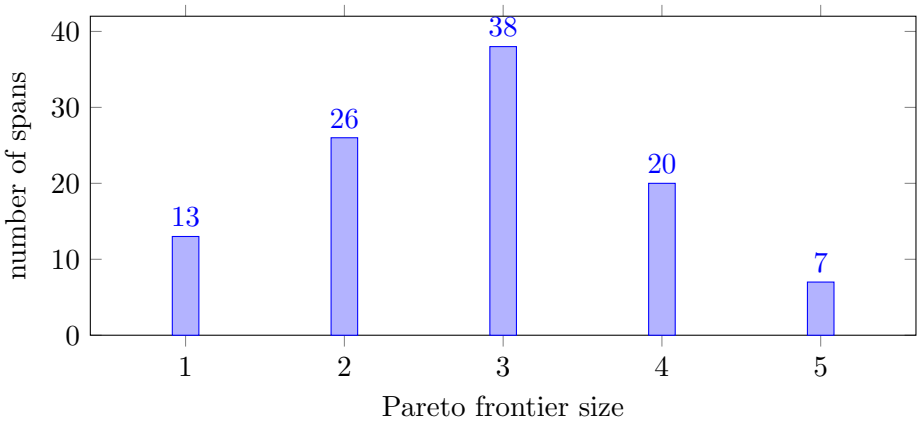



Figure 2: Distribution of Pareto frontier sizes $|\Pi(\sigma)|$ across the 103 adjacent-pair records (supplied [COMPUTED] datum, R275 dossier).

scripted audits are [COMPUTED], not formalized; R273's original T2 census-universe conclusion is FALSE ([COMPUTED] decisive: 51/294 Pareto instances have a witness whose reflection dual $(e', s)$ satisfies $e' < e$; 3937/5872 shapes' dual classes are absent from their span's census spectrum — the census spectrum is not reflection-closed, affecting 83% of Pareto instances; supplied datum, R275 dossier).

**Theorem 5** (Fiber-overlap; PROVED)**.** *Let $A$ be a Sidon set (Lean `IsSidon`: $a + b = c + d$ implies $\{a, b\} = \{c, d\}$ as multisets). Then:*

1. *(`autocorr_le_one`) for $t \neq 0$,*
$$|A \cap (A + t)| \leq 1; \tag{2}$$
2. *(`fiber_overlap_identity`) for all shifts $s, s'$,*
$$|(s - A) \cap (s' - A)| = |A \cap (A + (s - s'))|; \tag{3}$$
3. *(`fiber_overlap`, the Fiber Overlap Lemma) for $s \neq s'$,*
$$|(s - A) \cap (s' - A)| \leq 1. \tag{4}$$

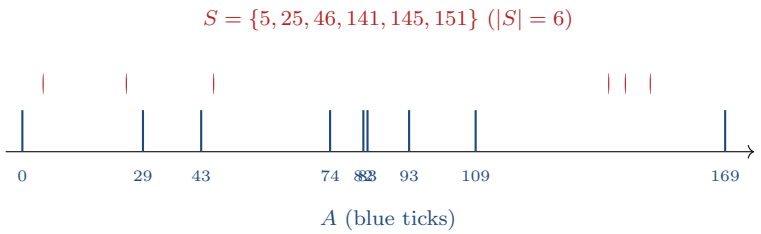



Figure 3: The six-hole witness of Proposition 1: the Sidon 9-shape (6) has $|E| = 0$ but only six $S$-holes (red), refuting the claimed $|S| \geq 7$ floor. Data from `r294_q20_verify.json` (supplied datum, R294 dossier).

The identity (3) is the bridge: fiber overlap *is* autocorrelation, and autocorrelation is $\leq 1$ for Sidon sets because two distinct representations of a shift would give two distinct pair-sum decompositions. The module additionally proves `f_iff_f1_f2` [PROVED, pure logic]: Conjecture F ($|E| + |S| \geq 2$ for all Sidon 9-shapes) holds iff both F1 ($|E| = 0 \implies |S| \geq 2$) and F2 ($|S| = 0 \implies |E| \geq 2$) hold. The 5,813,280-pair fiber audit found 0 mismatches of the identity, 0 violations of the $\leq 1$ bound, and 0 cases of autocorrelation $> 1$ (supplied [COMPUTED]$\times 2$ datum, R277 dossier).

**Theorem 6** (Top-interval sparsity, conditional; PROVED)**.** *For $\sigma \leq 91$, given $d \leq u$ with $u = 183 - \sigma$, $u - d = e$ (T2 equality, hypothesis `hT2`), and $e \geq 183 - 2\sigma$ (T1 E-floor, hypothesis `hT1`),*

$$d \leq \sigma, \tag{5}$$

*i.e. $|D \cap [\sigma + 1, 183]| \leq \sigma$.*

Theorem 6 (`top_interval_sparsity`) is [COMPUTED] on 377 universe shapes with $\sigma \leq 91$, 0 violations (supplied datum, R277 dossier), and [PROVED] as a Lean conditional. The condition is essential: T1 and T2 are [PROVED*] (R273), cited here with labels intact and never used as established fact. The Q-23 review scope is exactly the three fiber lemmas; the scripted audit and the $\lceil u/9 \rceil$ fiber-budget tightening application are not formalized.

**Proposition 1** (Q-20 refuted; COMPUTED)**.** *The claim "$|E(A)| = 0 \implies |S(A)| \geq 7$ for every Sidon 9-shape" is false. The exact witness (fresh code path `r294_q20_verify.py`, $9 \times 9$ pair-sum grid deduped, plain set loops; result JSON `r294_q20_verify.json`) is*

$$A = \{0, 29, 43, 74, 82, 83, 93, 109, 169\}, \quad \sigma = 169, \tag{6}$$

*a valid Sidon 9-shape ($|P| = 45$) with $E = \varnothing$ and*

$$S = \{5, 25, 46, 141, 145, 151\}, \qquad |S| = 6 < 7, \tag{7}$$

*$|D| = 262$, $|H| = 27$, $|J| = 14 \subseteq D$ (supplied datum, R294 dossier).*

Consequently Conjecture F's O1 branch is not closed by this route; F1/F2 remain [OPEN] (Q-11, Dot-owned, cited as an open dependency). R270's sweep re-analysis stands: 40 spans achieve true $|E| = 0$ (min $|S| = 7$ at $\sigma = 158$); 18 spans true $|S| = 0$ (min $|E| = 5$ at $\sigma \in \{116, 130\}$) (supplied datum, R277 dossier). We do not claim the refuted floor anywhere in this paper.

## 5 Discussion

**Significance.** The transfer suite isolates the abstract, machine-checkable core of the Pareto-transition analysis from its computational census. Theorem 2 shows the entire transition taxonomy is *local*: statuses are functions of down-set queries on the two spectra, so any dispute about

a transition reduces to a dispute about down-set data. Theorem 3 is a no-go theorem in the precise sense: no one may claim that A1–A3 *force* consecutive spectra to be distinct, to overlap, or to relate at all — the three models are explicit Lean-checked counterexamples to any such forcing claim. The fiber suite (Theorem 5) gives the march-graph framework its basic overlap law: fiber geometry is autocorrelation geometry, bounded by Sidonness itself.

**The refutation record.** Proposition 1 stays on the record because it changed the program's direction: the sharp $|S| \geq 7$ Pareto floor was a reviewer's claim, and its refutation by the six-hole witness closed the O1 branch of Conjecture F by this route. The R273 T2 census-universe conclusion was likewise falsified ([COMPUTED] decisive, 51/294 instances with dual $e' < e$), and T2′ in its per-shape form is DEFERRED pending formalization of the $(S, E)$ evaluator. The paper claims only what the Lean modules prove; the census remains [COMPUTED]×2 evidence, never a theorem.

**Limitations.** (i) Theorem 2 characterizes the status *function*, not the actual transitions: which statuses occur in the 103-pair census is a [COMPUTED] fact, not a theorem. (ii) Theorem 3's models are tiny singletons/pairs with identity reflection; they rule out axiom-level forcing but say nothing about which configurations the real $(S, E)$ evaluator can produce. (iii) Theorem 6 is conditional on T1/T2 [PROVED*] and the $\lceil u/9 \rceil$ fiber-budget application that motivates it is not formalized. (iv) The in-universe Pareto gap ($\min_\sigma \min_{(e,s)\in S(\sigma)}(e+s) = 11$ at $\sigma = 128$; corners $(0,0), (1,0), (0,1)$: 0 hits) and the boundary classes ($(5,0)@\{116,130\}$, $(0,9)@\{147,166,167,172\}$, etc.) are supplied [COMPUTED] datums, not theorems of this paper.

## 6 Conclusion

We have formalized, and proved in Lean, the $\sigma$-transfer core (Pareto membership as down-set membership, T5 transition locality, the T6 no-forcing theorem with three explicit models, the T7′/T8′ dual statements, the Pareto axiom) and the fiber-overlap suite (the overlap–autocorrelation identity, the $\leq 1$ bound, the F $\iff$ F1∧F2 reduction, and conditional top-interval sparsity), all [PROVED] with zero sorrys and standard-only axioms on the bakkie full-library build. The census numbers that motivate the work are [COMPUTED]×2 and stay labeled as such; the Q-20 sharp-floor claim is recorded as refuted by the exact six-hole witness; the per-shape T2′ and the F1/F2 fragments remain [OPEN]. The natural next steps are formalizing the $(S, E)$ evaluator (which would unblock per-shape T2′), and the Dot-owned F1/F2 fragments of Conjecture F.

## 7 References

## References


[1] P. Erdős and P. Turán, *On a problem of Sidon in additive number theory, and on some related problems*, J. London Math. Soc. **16** (1941), 212–215. (Cited as given among the program's classical Sidon references.)

[2] OpenAI, *github.com/openai/math* catalogue, 719 manuscripts fetched 2026-10-10, 0 "Sidon" hits (cited as given, R273/R275 novelty check).

[3] D. Blair, *Unsolved Math — Round 275: $\sigma$-transfer*, program dossier, 2026-10-10. (Supplied datum: transition taxonomy, census, T5/T6/T7′/T8′ audits, Q-22 scoped verdict.)

[4] D. Blair, *Unsolved Math — Round 277: fiber overlap*, program dossier, 2026-10-10. (Supplied datum: fiber identity audit, 5,813,280 pairs, sparsity audit, Q-23 verdict.)

[5] D. Blair, *Unsolved Math — Round 294: Q-20 refutation*, program dossier, 2026-10-10. (Supplied datum: six-hole witness, `r294_q20_verify.json`, Q-23 scope record.)

[6] D. Blair, *Unsolved Math — Round 298: queue hardening*, program dossier, 2026-10-10. (Supplied datum: md5 sync records, `sorryAx` fix history.)

[7] D. Blair, *Unsolved Math — Round 310: full-library build gate*, program dossier, 2026-10-11. (Supplied datum: bakkie `lake build` "Build completed successfully (17936 jobs)", exit 0, 0 errors, 0 sorrys; [Q-22]/[Q-23] label moves.)